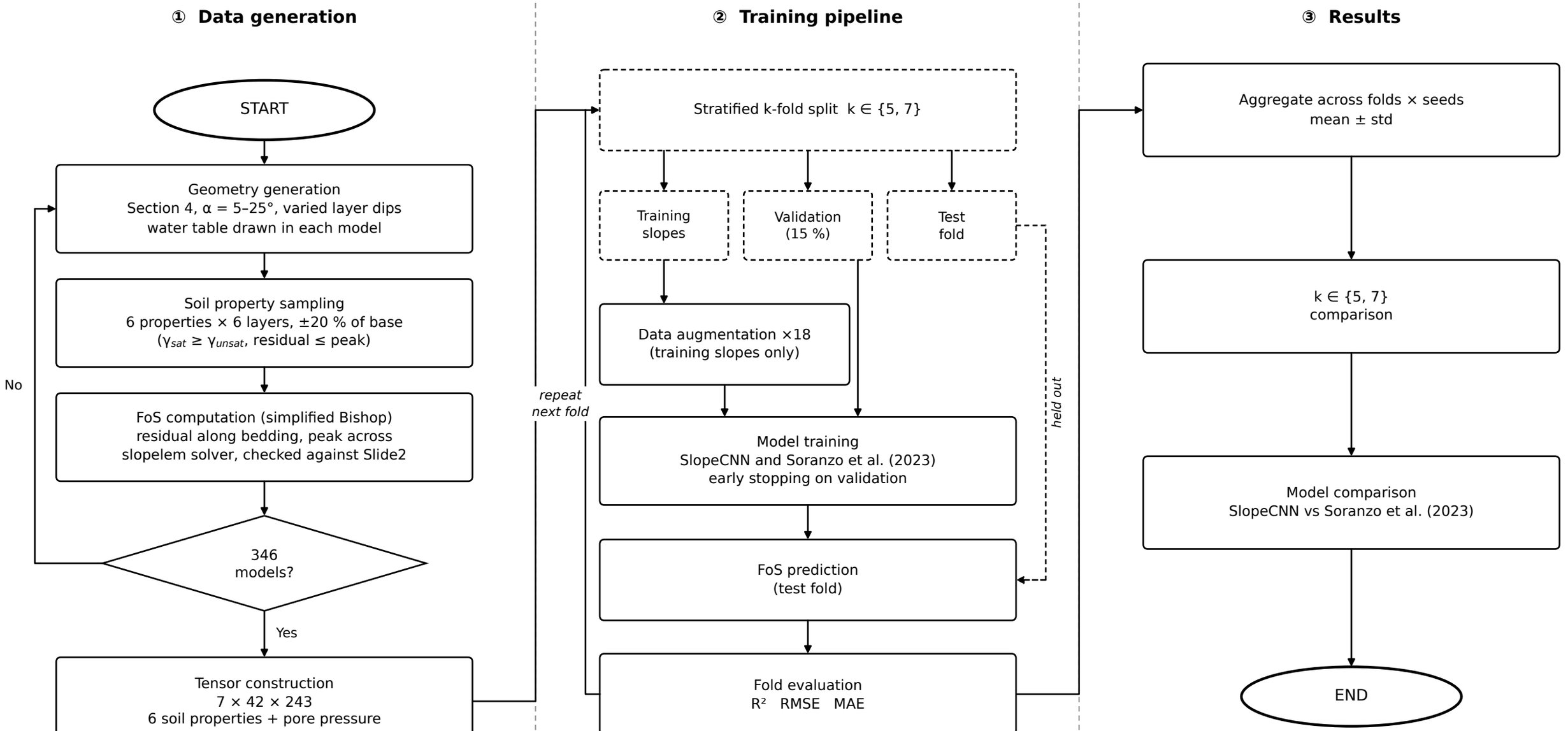
① Data generation
START
Geometry generation
Section 4, α = 5–25°, varied layer dips
water table drawn in each model
Soil property sampling
6 properties × 6 layers, ±20 % of base
(γsat ≥ γunsat, residual ≤ peak)
FoS computation (simplified Bishop)
residual along bedding, peak across
slopelem solver, checked against Slide2
346
models?
No
Yes
Tensor construction
7 × 42 × 243
6 soil properties + pore pressure
② Training pipeline
Stratified k-fold split k ∈ {5, 7}
Training
slopes
Validation
(15 %)
Test
fold
Data augmentation ×18
(training slopes only)
Model training
SlopeCNN and Soranzo et al. (2023)
early stopping on validation
FoS prediction
(test fold)
Fold evaluation
R² RMSE MAE
repeat
next fold
held out
③ Results
Aggregate across folds × seeds
mean ± std
k ∈ {5, 7}
comparison
Model comparison
SlopeCNN vs Soranzo et al. (2023)
END